PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 26 1996

Relaxation methods allow the study of fast reactions. A reaction mixture at equilibrium is disturbed by a pressure, thermal, or electrical shock and allowed to return to equilibrium. The restoration is always first-order for small displacements from equilibrium. For the following generic reaction the the relaxation time (τ) can be written in a very simple form as shown.

$$aA + bB \underset{k_2}{\overset{k_1}{\rightleftharpoons}} cC + dD$$

$$1/\tau = k_1 [A_{eq}]^a [B_{eq}]^b (a^2/[A_{eq}] + b^2/[B_{eq}]) + k_2 [C_{eq}]^c [D_{eq}]^d (c^2/[C_{eq}] + d^2/[D_{eq}])$$

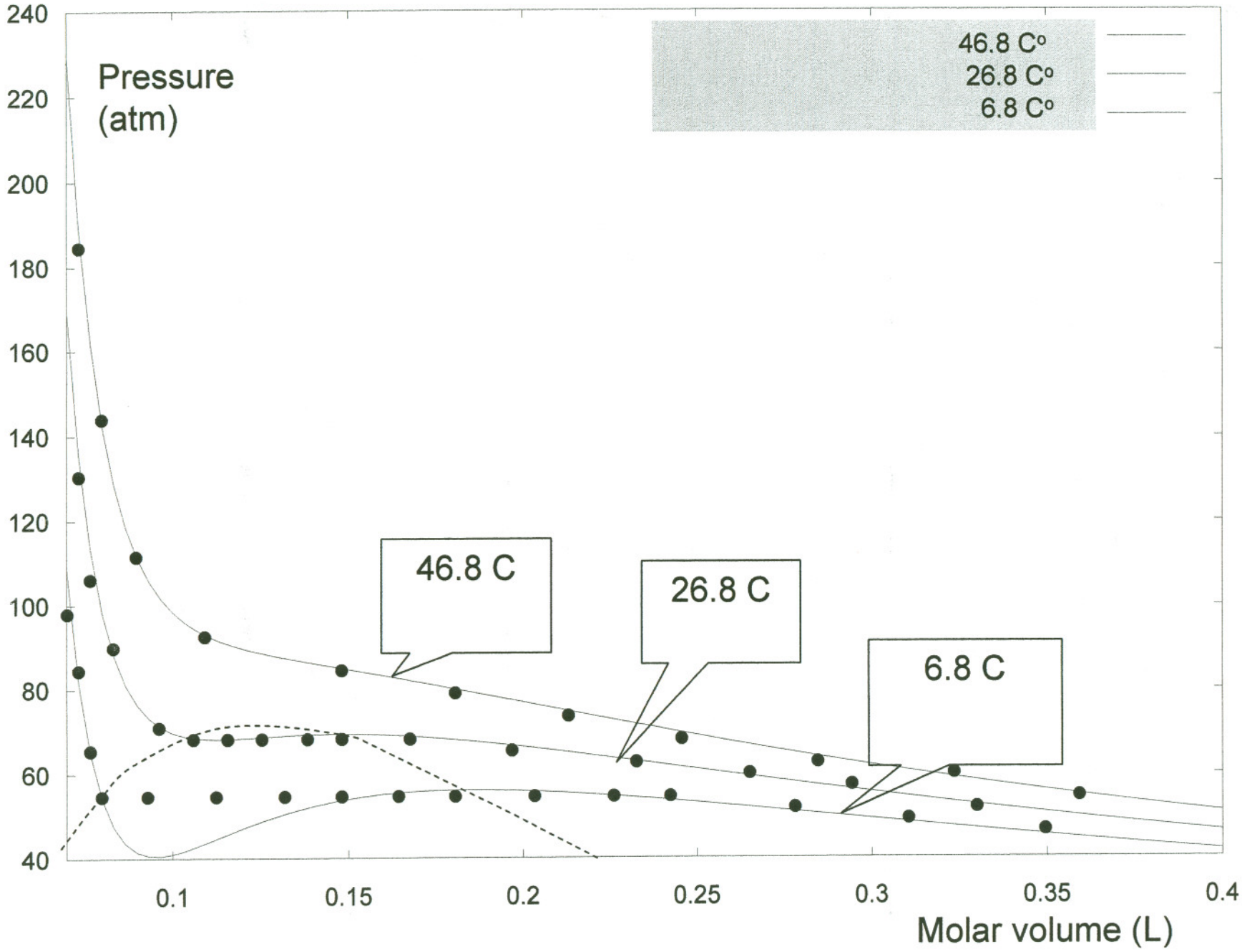
Using the pressure jump technique, the data shown in the table below were obtained for the reaction:

$$Ni(NCS)^+ + NCS^- \underset{k_2}{\overset{k_1}{\rightleftharpoons}} Ni(NCS)_2$$

Determine the value of k_1 and k_2. You may assume that $[Ni(NCS)^+]_{eq} = [(NCS)^-]_{eq}$

You may also assume that K_{eq} is **very** large.

$C_{Ni(NCS)_2}$, M	0.001	0.002	0.005	0.010	0.025	0.05	0.10
τ, ms	4.08	3.74	2.63	1.84	1.31	0.88	0.67


Pressure
(atm)
46.8 C°
26.8 C°
6.8 C°
46.8 C
26.8 C
6.8 C
240
220
200
180
160
140
120
100
80
60
40
0.1
0.15
0.2
0.25
0.3
0.35
0.4
Molar volume (L)